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Wind Turbine Failure Prediction Using SCADA Data

1 Overview

Wind turbines are critical assets in modern renewable energy systems, yet unexpected component failures lead to costly downtime, expensive repairs, and lost power generation. Traditional maintenance strategies are largely reactive or schedule-based and do not fully exploit the rich sensor data already collected by turbine monitoring systems. More reliable early-warning systems are needed to detect abnormal operating conditions before failures occur.

Project objective. This project aims to develop a supervised machine-learning model that predicts whether a wind turbine is likely to experience a fault within a short future time window (e.g., 24–48 hours) using historical SCADA telemetry data. The main goals are:

- Formulate the problem as a binary classification task (failure vs. normal operation).
- Identify sensor variables that are most indicative of impending failures.
- Compare multiple machine-learning models for predictive accuracy and interpretability.
- Evaluate performance using metrics relevant to engineering decision-making (e.g., recall of failure events).

2 Dataset Description

Source. The dataset used in this project is publicly available on Kaggle and Zenodo:

Kasimov, A. (2024). *Wind turbine SCADA data for early fault detection* [Data set]. Zenodo. https://zenodo.org/records/10958775

Key features. The dataset contains time-stamped measurements from multiple turbines, including:

- Environmental variables (e.g., wind speed, ambient temperature)
- Mechanical variables (e.g., rotor speed, generator speed)
- Electrical variables (e.g., power output, voltage, current)
- Thermal variables (e.g., gearbox and bearing temperatures)
- Operational indicators and fault/anomaly labels

Size and structure. The data consist of multivariate time-series records collected at regular intervals from multiple turbines operating across several wind farms, comprising thousands of observations. Each row represents a snapshot of turbine operating conditions at a given time, with an associated status label.

Limitations and challenges. Potential challenges include class imbalance (failures are rarer than normal operation), missing or noisy sensor values, and uncertainty in defining the optimal prediction horizon for near-term failures.

3 Proposed Methodology

Preprocessing and exploration. The dataset will be cleaned by handling missing values, removing or capping extreme outliers, standardizing numerical variables where appropriate, and ordering observations chronologically. Exploratory data analysis (EDA) will be conducted to compare sensor behavior during normal operation versus pre-failure periods and to examine correlations among variables.

Feature engineering. Temporal features such as rolling means, standard deviations, and rates of change will be generated to capture degradation trends. Basic feature selection will be applied using correlation analysis and model-based importance scores. Dimensionality reduction (e.g., PCA) will be considered if the feature space becomes excessively large.

Models. The following supervised learning algorithms are planned:

- Logistic Regression
- Random Forest
- Gradient Boosting

A time-aware train/validation/test split will be used to avoid information leakage from future observations.

Evaluation metrics. Model performance will be assessed using accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). Particular emphasis will be placed on recall for the failure class, as missed failures have high operational cost.

4 Expected Outcomes

The project is expected to produce a validated machine-learning pipeline capable of estimating near-term turbine failure risk from SCADA data. Results should highlight which sensor variables provide the strongest early-warning signals and demonstrate how predictive models can support condition-based maintenance planning. Anticipated challenges such as class imbalance and sensor noise will be addressed through appropriate resampling techniques, feature engineering, and careful evaluation. Overall, the project will illustrate how data mining methods can improve the reliability and cost-effectiveness of renewable energy infrastructure.